

Code-Layout, Readability and Reusability

Date	09 July 2026
Team ID	SWTID-2026-5556
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper 4-space indentation used throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into src, model, dataset, templates, static folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables like applicant_income, credit_score, prediction_result are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as preprocess_data(), train_model(), predict_loan() are meaningful.
5	Code Comments	Inline and block comments explain logic	Yes	Important functions are commented for easy understanding.
6	Modular Design	Code is split into reusable functions/modules	Yes	Preprocessing, training, prediction, and Flask app are implemented separately.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code removed by using reusable functions.

8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling are implemented where required.
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Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	preprocessing.py	Python	Data cleaning and feature preprocessing	High

2	train_model.py	Python	Model training	High
3	predict.py	Python	Loan prediction	High
4	app.py	Flask	Web application	High
5	model.pkl	Machine Learning	Prediction module	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well organized project folders and files.
Readability	5	Easy to understand and maintain
Reusability	5	Functions and modules can be reused
Documentation / Comments	4	Adequate comments included.
Overall Score	5	Clean, modular, and maintainable code.